

Pravesh Pandey

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PROFESSIONAL SUMMARY

Software Development Engineer with 3 years of experience building low-latency, high-throughput backend and distributed systems for Amazon Alexa. Expertise in Java, Python, C++, AWS, microservices, event-driven architecture, concurrency, performance engineering, and production reliability. Delivered systems operating at 6K–40K TPS, reduced latency by up to 550 ms, and scaled an evaluation platform to 60+ teams.

EXPERIENCE

Amazon

Bengaluru, India

Software Development Engineer

May 2024 – Present

- Architected a deterministic fast path with publish-subscribe orchestration and a 61 GB distributed cache; served 6K TPS across 11 locales, cut latency 400–550 ms, and improved quality by 140 basis points.
- Engineered locale-specific finite-state transducers (FSTs) for Pattern Match hotfixes across 18 Classic Alexa and Alexa+ locales, reducing artifact size by 58% and latency from 40 ms to 25 ms.
- Resolved incidents across five services, including FST corruption at 40K TPS and a multi-Region outage; added alarms and a five-minute runbook to accelerate mitigation.
- Automated Operational Readiness Reviews with a five-agent orchestrator and production Model Context Protocol (MCP) server; implemented nine safety controls and reduced months of manual effort to 11 agent-hours.
- Scaled LangForge from a one-week proof of concept to adoption by 60+ Alexa teams, integrating three partner systems and resolving 15+ edge cases.
- Reduced evaluation failures from roughly 60% to near zero and runtime from 18 hours to 12 minutes, saving 160–200 engineering hours per cycle.
- Modernized nine microservices from JDK 8 to JDK 17, simplifying maintenance and reducing compute costs by 15%.

nFolks Data Solutions

Remote, India

Software Engineer

Oct 2023 – Apr 2024

- Developed a multithreaded Python data-processing framework that cut database retrieval time by 68%; optimized pandas pipelines and deployed Flask REST APIs backed by SQL through Jenkins CI/CD.

Phyt Health

Pune, India

Software Development Intern

Jan 2022 – Mar 2022

- Created an AI-guided fitness application using React, Node.js, TypeScript, TensorFlow, and C++, delivering real-time posture correction, exercise tracking, and Twilio video to 500+ pilot users with 4.2/5 feedback.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, JavaScript, TypeScript, C#

Backend & Distributed Systems: Microservices, REST APIs, GraphQL, event-driven and publish-subscribe architecture, distributed systems, multithreading and concurrency, low-latency and high-throughput design, high availability, fault tolerance, distributed caching

Cloud & DevOps: Amazon Web Services (AWS: ECS, Fargate, Lambda, S3, DynamoDB, CloudWatch, KMS, IAM, CDK, CloudFormation, Step Functions, Kinesis, ElastiCache, ALB, Auto Scaling), Azure, Docker, Jenkins, Git, CI/CD

Frameworks & Data: Node.js, React, Flask, .NET, MySQL, MongoDB, Redis, pandas, NumPy, TensorFlow

Foundations & AI: Data structures & algorithms, system design, performance and capacity engineering, load testing, stress testing, resilience testing, observability, monitoring, incident response, root cause analysis (RCA), generative AI, large language models (LLMs), multi-agent systems, Model Context Protocol (MCP)

EDUCATION

Motilal Nehru National Institute of Technology

Uttar Pradesh, India

Bachelor of Technology

Jun 2019 – Jun 2023

AWARDS AND CERTIFICATIONS

Awards: Alexa AI AIDo Frugal Frontrunner Award (2025), Amazon Extra Mile Award (2025), MNNIT Project Exhibition Winner (2023)

Certifications: MongoDB Specialization — MongoDB, Inc. (2023)